Executable Verification and Analytic Repair in LLM-Guided Quantum Circuit Search: What a Controlled Benchmark Does, and Does Not, Show

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Abstract

We ask what an executable verifier and an analytic repair step actually change when a large language model (LLM) turns a *symmetry hypothesis* into an *executable* quantum predictor, under matched evaluation budgets. The mathematical tools we use (group twirls, channel and predictor equivariance residuals, an orthogonal-projection risk identity) are established; the object of study is behavioural and measurable: the gap between *proposing* a transformation, *implementing* it, and *performing*. This version was written against a referee report, and every change below is traceable to a frozen revised protocol, a corrected analysis, or a bibliographic record. Four things survive measurement. (i) A self-audit of our own prior LLM-evolutionary study: its selection score used test data, one architecture-wide equivariance claim is false (measured channel defect 0.0597), and significance was computed at proposal rather than run level. (ii) A causal repair experiment on a ring benchmark whose replication units, controls and floors are stated explicitly: at η=0 the target is direction-free, so the previous 24-row record contains 8 units, not 24 (recomputed: interval $[-0.00217, -0.00002]$, sign-flip p=0.125); the redesigned experiment with 24 genuine η=0 units finds that the repaired arm beats the original (-0.00134, Holm p=0.043) but *not* the gate-count-matched duplicate control (-0.00060, CI $[-0.00135, +0.00019]$), so the symmetry-specific benefit is not established, while the η=1 cost of enforced invariance is large, robust and budget-increasing ($+0.03605$, CI $[+0.03134, +0.04087]$ at 200 steps, rising to $+0.057$ at 1800). (iii) In the frozen 120-run search study, the endpoint that had been omitted — inference-time Reynolds averaging of the deployed predictor — is the one effect that is consistent and large (-0.0486, CI $[-0.0708, -0.0284]$, 7/12 runs eligible on `graphs_n6`), whereas verifier *feedback* changes development-selected risk by an undetected amount; the same averaging applied to non-LLM arms helps equally, so it is the analytic tool, not the LLM, that carries the gain. (iv) Verification as implemented does not identify the group: 10 of 299 graph proposals carried the task's group and 289 wrong-group proposals passed the label-preservation gate — which is expected, because every subgroup of a label symmetry passes it; it is a soundness filter, not an identification test. A hand-built equivariant architecture beats every search arm on the ring ($+0.0060$ to $+0.0097$ worse for the arms), and a one-threshold classical rule beats every arm on graphs (dev risk 0.351 vs 0.538–0.566). All endpoints are labelled development, sealed or fresh; we do not claim held-out generalisation from development risk, we do not claim quantum advantage, and we record two provenance defects in our own earlier run records rather than quietly fixing them.

1 Introduction

Symmetry is a first-class inductive bias in quantum machine learning (QML): a circuit whose blocks commute with a representation of the data's symmetry group, combined with an equivariant encoding, an invariant initial state and a compatible readout, yields invariant predictions. A tempting inference is that an LLM asked to design such circuits will both *state* the right symmetry and *build* it, and that giving the model an executable check will close the gap.

This paper treats that inference as measurable, and separates three levels that casual "symmetry-aware LLM" claims conflate: (i) **proposing** a transformation (a stated group hypothesis); (ii) **implementing** it (the circuit actually commutes with the representation, at the parameters that were trained); and (iii) **performing** (the trained predictor is useful on data that no part of the pipeline adapted to). Our thesis is that the interesting, testable phenomenon is the *gap* between these levels, and that measuring it requires an endpoint for each level — not one endpoint read as three.

How this version differs. The manuscript was reviewed and found to contain identifiable validity problems: duplicated experimental units in the headline repair result, a search endpoint that did not measure the intervention it credited, generalisation language attached to development data, an unjustified "differs only in symmetry" claim, an "if and only if" claim, a mis-stated risk floor, uncalibrated p-values, a prose/table contradiction, and two materially wrong references. All were checked against the code and the primary records; all were real. Section 6 states the revised protocol frozen before any new result existed, and Appendix C records the dated chain, the model identity, the compute budget, and two provenance defects we found in our own earlier run records.

Contributions.

1. **A self-audit** (§4) of the prior in-house study this project continues [9]: test data inside the selection score, a refuted architecture-wide equivariance claim, and proposal-level significance.
2. **A causal repair experiment with explicit units, controls and floors** (§5, §7.1): $y_\eta = h + \eta b$; an operator-matched asymmetric control in addition to a gate-count-matched one; a second asymmetric control; optimiser-budget sensitivity; and a function-space evaluation on orbit-completed data that separates the population floor, its Monte-Carlo estimate, and the exact finite-orbit floor.
3. **A reanalysis of the frozen 120-run search study that restores the missing endpoint** (§6, §7.2): the paired effect of inference-time averaging on the frozen candidate, the deployed-pipeline contrast, coverage of the intervention, paired reliability endpoints, and verifier selectivity measured against the true group.
4. **Baselines that the LLM story has to beat** (§7.5): a constructive equivariant architecture at the same fit budget, a task-aware classical rule, the invariant oracle, and the same repair tool applied to non-LLM search.

2 Related work, re-verified

Every entry of §2 was checked against a primary record (arXiv API, Crossref, PMLR); all eight had at least one material error, and the corrected list is in the bibliography with the audit in `research/paper/REFERENCES_AUDIT.md`. Symmetry exploitation in variational QML is established

Table 1: Contribution-to-evidence table. "Evidence" names the executed artefact; "Does not show" is the boundary we refuse to cross.

Claim	Evidence	Does not show
Prior selection leaked test data	`research/audit.py` executes the evaluator's own AST; changing every test-derived field leaves the v2 score unchanged	that the historical task had no signal; it had a density shortcut
Enforcing equivariance changes risk causally	WP-REPAIR-2, 8 directions × 24 optimiser seeds, gate-count- *and* operator-matched controls (Table 4): robust cost at $\eta>0$, *no* established symmetry-specific benefit at $\eta=0$	an "iff" over all mismatches; a symmetry-only effect (families differ in parameter-to-function map and optimisation geometry)
Inference-time averaging helps the deployed predictor	E-S1: -0.0486, CI $[-0.0708, -0.0284]$, 7/12 eligible runs, graphs; E-S5b: same tool helps every arm (§7.2)	that the LLM found the group; coverage-conditional, not intention-to-treat
Verifier feedback improves search	**Not established**: primary difference undetected on both families, CIs wider than the prespecified equivalence margin	equivalence; "clean null" is not a finding we claim
Verification identifies the symmetry group	**Refuted as implemented**: 10/299 correct groups on graphs, 0/192 on ring; 289 wrong-group proposals pass label preservation	that no verifier could identify a group; our gate is a soundness filter
LLM search beats simple alternatives	**Not established**: constructive equivariant beats all arms on ring; edge-count threshold beats all arms on graphs	quantum advantage of any kind

work [1]; equivariant unitary layers and channels, with constructive prescriptions, are general and known [2]; permutation-invariant QML for graph connectedness is directly occupied territory [3]; symmetry alone does not determine a useful architecture [4]; averaging-operator generalisation benefits, including projection identities of the kind we use, are known [5]; enforcing, discovering and promoting symmetry by Lie-derivative methods is known [6]; finite-measurement task-level symmetry inference with projection bias is the closest conceptual neighbour [7]; and the LLM-evolutionary harness we build on is ShinkaEvolve [8]. The bandit ensemble that selects a proposer per generation is *not* from [8]: it is a modification made by our own prior study [9], and we attribute it there.

The mathematical tools are therefore not the contribution, and we make no priority claim. What we intend to contribute is empirical and negative-leaning: a measurement, under matched budgets, of *where* executable verification changes behaviour in LLM-guided quantum circuit design — finding that the analytic repair step carries the gain while the group-identification step fails in a way that is predictable from the group theory.

3 Preliminaries: symmetry diagnostics and their assumptions

Let P_g be a specified unitary representation of a finite group G on $\mathbb{C}^d$ and U a circuit unitary at fixed numerical parameters. The *strict* residual

$$D_U(g) \;=\; \frac{\|U - P_g U P_g^\dagger\|_F^2}{2d} \in [0,2] \tag{1}$$

vanishes iff U commutes with P_g. For physical unitary channels we use the *channel* residual

$$D_{\text{chan}}(g) \;=\; 1 - \left|\mathrm{Tr}(U^\dagger P_g U P_g^\dagger)/d\right|^2 \in [0,1], \tag{2}$$

which vanishes exactly when the conjugation differs by a global phase (e.g. Z anticommutes with X: strict residual 2, channel residual 0). Adjacent transpositions generate S_n; zero residual on all generators certifies exact symmetry *at the sampled parameters*. A random-parameter zero is evidence about a family, not a proof over a family; where a proof exists we give it (Appendix A).

3.1 From block equivariance to predictor invariance

Block equivariance alone does not give an invariant predictor. Writing the expanded circuit as $V(x) = B_L\, U_{\text{enc}}(x) \cdots B_1\, U_{\text{enc}}(x)$ with readout M, $f(x) = \langle 0\cdots 0|V(x)^\dagger M V(x)|0\cdots 0\rangle$ is G-invariant when

1. **encoding equivariance:** $U_{\text{enc}}(gx) = P_g\, U_{\text{enc}}(x)\, P_g^\dagger$ (the ring encoding applies $\mathrm{RY}(x_i)$ on wire i, so a site permutation is implemented by P_g; the graph encoding applies $\mathrm{RZZ}_{\pi/2}$ on the edge indicated by bit k, and S_n acts on edges);
2. **invariant preparation:** $|0\cdots 0\rangle$ is fixed by P_g;
3. **block equivariance:** each block commutes with P_g;
4. **invariant readout:** $[M, P_g] = 0$ for all g — we use the fixed readout $M = \frac{1}{n}\sum_i Z_i$ and do *not* train the readout, because a learned readout erases the bias.

Conditions (1), (2) and (4) are structural and checked once; condition (3) is what the residual measures. Because a residual on the block is not a certificate on the predictor, we additionally measure predictor invariance directly, on all group images of every evaluation row: with f fitted and $\mathcal{T}_G f$ its Reynolds average over the orbit, $\mathbb{E}\big[(f - \mathcal{T}_G f)^2\big]$ is reported per arm (§7.1, Table 5); it is 0 to roundoff for the repaired arm and $O(10^{-2})$ for the asymmetric arms.

3.2 The projection identity, and the floor, with their measures stated

The Reynolds operator $\mathcal{T}_G(U) = \frac{1}{|G|}\sum_g P_g U P_g^\dagger$ is an orthogonal projection for the Hilbert–Schmidt inner product, so the mean strict residual over a uniform group equals $\|U - \mathcal{T}_G(U)\|_F^2/d$. For predictions on inputs, let μ be the input distribution and let $\bar\mu = \frac{1}{|G|}\sum_g g_\# \mu$ be its orbit average. μ is *invariant* iff $\bar\mu = \mu$.

Proposition 1 (risk decomposition on an invariant input measure)**.** *Let μ be G-invariant, $y = h + \eta b$ with $\mathcal{T}_G h = h$ and $\mathcal{T}_G b = 0$, and $R(f) = \mathbb{E}_{x\sim\mu}(f(x) - y(x))^2$. Then for every f,*

$$R(f) \;=\; R(\mathcal{T}_G f) + \mathbb{E}\big[(f - \mathcal{T}_G f)^2\big] - 2\,\mathbb{E}\big[(f - \mathcal{T}_G f)(y - \mathcal{T}_G y)\big], \tag{3}$$

so the cross term vanishes when y is invariant ($\eta = 0$), giving $R(f) = R(\mathcal{T}_G f) + \mathbb{E}[(f - \mathcal{T}_G f)^2] \geq R(\mathcal{T}_G f)$, and for every invariant *f, $R(f) = \mathbb{E}[(f-h)^2] + \eta^2\mathbb{E}[b^2] \;\geq\; \eta^2\mathbb{E}_\mu[b^2]$.*

The proof is the orthogonality of the projection plus $\mathbb{E}[g\,b] = \mathbb{E}[g\,\mathcal{T}_G b] = 0$ for invariant g, which uses the invariance of μ essentially. The sign matters and an earlier draft of this paper had it wrong: with an invariant target the post-hoc projection can only *help* ($R(f) \geq R(\mathcal{T}_G f)$), but when the target breaks the symmetry the cross term dominates and $R(f) - R(\mathcal{T}_G f) < 0$ — measured, -0.013 to -0.018 at η=0.5 and -0.033 to -0.047 at η=1 for the asymmetric arms, i.e. projecting an asymmetric fit onto the invariant subspace *hurts* exactly where invariance is wrong. The full identity holds to 3.7×10^{-16} over all 2340 fits of WP-REPAIR-2 (5.5×10^{-18} at η=0). In our benchmark $\mu = \mathrm{Unif}[-1, 1]^6$ with the site-permutation action, a product of identical marginals, hence permutation-invariant, so the hypothesis holds. Three quantities must not be conflated, and the first version of this paper did conflate them:

- the **population floor** $\eta^2 \mathbb{E}_\mu[b^2]$ (here η^2 exactly, because b is normalised on the training split);
- a **Monte-Carlo estimate** on sampled orbit representatives, which has sampling error (on our development split it is 0.866–1.059 across breaking directions at η=1, mean 0.948 — the first version of this paper quoted one direction's value, 0.866, as if it were the floor);
- the **exact finite-orbit floor** on the orbit-completed evaluation set (each representative replaced by all $|G|$ images, uniform weights), which is exact for the sampled orbits and is the Rao-Blackwellised version of the second.

A sample containing one representative per orbit admits invariant interpolants of its labels, so no non-vacuous empirical lower bound exists on the representative sample alone; that is why the orbit-completed set, with declared weights, is what we use for floor accounting.

4 Self-audit of the prior study

The audited study is *an earlier version of this project by the same author*: our prior study [9], archived in this repository as `README.md` at git commit `13ea13eb76af2f587c508aefde81b9dd026b3ce1` with run artefacts under `experiments/transfer-sn/` and `analysis/`. We state this explicitly because auditing one's own prior work changes what the audit can and cannot establish: we are competent to inspect the code, and we are also the party with an incentive to soften the findings. Four findings bound what the old numbers can mean; each is reproducible with `python3 research/audit.py`, which executes the historical evaluator's own abstract syntax tree rather than a paraphrase of it.

Test leakage in selection. The selection score used a train–*test* gap and the public feedback disclosed test accuracy and loss. The historical "test" set measured an adaptively reused development set and cannot establish held-out generalisation.

A refuted equivariance claim. The hand-designed baseline was claimed to be permutation-equivariant for all parameters. Its one-direction CRZ layer is not: at a documented parameter setting its mean adjacent-generator channel defect is 0.0597, and a bidirectional half-angle repair reduces it to 1.5×10^{-14}. A separate ordered-CRX counterexample shows that covering every unordered pair does not guarantee symmetry, because non-commuting controlled rotations are order dependent.

Non-independence. 563 selected proposals are not 563 independent experiments: they share ancestry, data and adaptive selection, yet significance was computed proposal-by-proposal.

A trivial shortcut. A single edge-count threshold fitted on training data alone attains 83.0% historical validation accuracy. In the fresh benchmark used here the same rule reaches 91.2% development accuracy and risk 0.351 (§7.5), which is better than every search arm; density, not structure, carries much of the signal.

5 WP-REPAIR: a causal repair experiment with explicit units

Targets. On the ring benchmark (inputs $x \in \mathbb{R}^6$, group D_6 of order 12 acting by site relabelling), h is the dataset's ground-truth *invariant* oracle $h(x) = \frac{1}{n}\sum_i x_i x_{i+1}$ and, for a frozen random direction v, $b(x) \propto \big(v\cdot x - \mathcal{T}_G(v\cdot x)\big)$ is the group-centred remainder of a linear functional, normalised so $\mathbb{E}_{\text{train}}[b^2] = 1$. Targets are $y_\eta = h + \eta b$, $\eta \in \{0, 0.5, 1\}$. h and b are functions of the *input*, never produced by a compared ansatz, and the Reynolds projection is computed by applying the explicit group closure to each input (the dataset's orbit labels are singletons and cannot be used).

Replication units (the fix). At $\eta = 0$ the target does not contain b, so the breaking direction is a nuisance index, not a replication: the unit is the *optimiser seed*. At $\eta > 0$ the unit is the (direction, optimiser seed) pair, with direction a blocking factor. Analysis code counts rows and units separately and refuses to resample rows: the first version's "24 paired draws" at $\eta = 0$ are 8 units written three times, and Table 2 reports what changes when they are collapsed. We use 8 directions and 24 optimiser seeds, so $\eta = 0$ has 24 genuine units. Three breaking directions cannot support a population claim about arbitrary target mismatches; eight support a within-benchmark statement, and the cluster bootstrap over directions is reported with that caveat.

Arms. All arms train three block parameters $\{a, b, c\}$ with the fixed symmetric readout of §3.1.

- `original`: one-direction CRZ, 15 gates, asymmetric;
- `repaired`: bidirectional shared-angle CRZ, 30 gates, exactly equivariant (analytically, Appendix A);
- `control_dup`: same-direction CRZ twice, 30 gates, asymmetric;
- `control_rand`: per-pair direction from a fixed seeded rule, 30 gates, asymmetric;
- `control_matched`: `control_dup` with the initial angle halved, so its initial *operator* is identical to `original`'s.

What equal gate counts do and do not buy. `control_dup` is the `original` interaction at angle $2b$ (verified to 10^{-14}, Appendix A), so it matches gate count and reachable function family but *not* optimisation geometry, and identical numerical initialisations do not match initial operators: the measured initial-operator distance is 1.37 for `control_dup` and 0 for `control_matched`. We therefore describe the contrast as "the effect of these particular circuit interventions under this training procedure" and never as a symmetry-only effect. Optimiser-budget sensitivity (200/600/1800 steps) is reported separately.

Function-space estimand. For each fitted f we also evaluate f and $\mathcal{T}_G f$ on the orbit-completed development distribution (uniform over 12 group images $\times$ representatives), reporting $R(f) - R(\mathcal{T}_G f)$, $\mathbb{E}[(f - \mathcal{T}_G f)^2]$ and the cross term of Proposition 1. This is a deterministic statement about one predictor; it is not a statement about retrained architectures, and we do not use the former as a theorem about the latter.

6 WP-SEARCH: estimands, endpoints and statistics

The frozen record is the 120-run confirmatory study: five arms (`random`, `mutation-ea`, `llm-fitness`, `llm-verify`, `llm-verify-repair`) at identical evaluated-candidate budget B=12, identical seed-derived initial populations, identical per-candidate training budget (Adam, 120 steps, 256 training rows), every invalid proposal retained in the denominator, one model treatment (*Qwen3.8-Flash-Next* at reasoning effort `none`; identity pinned in Appendix C), 12 paired seeds per family.

Endpoints, named. *Development risk* is risk on the validation split of the development artefact; that split is used by candidate selection, by the search feedback loop and by the verifier, so it is not held out in any confirmatory sense. *Sealed test risk* is read once, after the protocol and analysis code were committed. *Fresh holdout risk* (ring only) uses a holdout generated after the freeze with zero canonical-orbit overlap with any development or sealed-test row. The first version of this paper wrote "held-out risk" for development numbers; that wording is removed everywhere.

Estimands. E-S1: paired effect of inference-time Reynolds averaging on the *same* frozen dev-selected candidate, reported with coverage (eligible / attempted / fired) because it is coverage-conditional and not intention-to-treat. E-S2: the deployed pipeline (averaged predictor where the repair fired, candidate otherwise). E-S3 (primary): `llm-verify` minus `llm-fitness` development-selected risk. E-S4: paired reliability endpoints (invalid-proposal rate, hypothesis-present rate, verifier false-accept rate against the true group, nonconstant-output check). E-S5: development risk inside edge-count bands of `graphs_n6` — the intervention the "density shortcut" explanation requires. E-S5b: the same averaging tool applied to every arm's selected predictor using the task's *declared* group, i.e. a non-LLM pipeline given the same tool. E-S6: matched-budget baselines.

Statistics. Percentile bootstrap of the mean paired difference over *units* (20,000 resamples); p-values from the null-centred (shifted) bootstrap plus a sign-flip test, whose assumption — the null distribution of paired differences is symmetric about 0 — is stated wherever it is used; simulated p-values carry a floor and are never reported as 0; leave-one-unit-out ranges accompany every primary; Holm correction within declared families only (A: the two primaries; B/C: the repair contrasts at η=0/η=1; D: secondary search contrasts per family and endpoint; E: the breaking-target study). A confidence interval containing 0 is reported as "difference not detected", never as "no effect", "unchanged" or "clean null". Practical equivalence is claimed only when a whole interval lies inside the prespecified margin $\delta = 0.05\,\mathrm{Var}_{\mathrm{dev}}(y)$ (0.03274 for graphs, 0.00097 for ring).

7 Results

7.1 Repair: corrected units change the headline

Table 2 resolves the referee's points 1, 5 and 8 on the existing record. The η=0 benefit shrinks to a borderline interval with sign-flip p=0.125, and nothing in the η=0 family survives multiplicity (Holm p=0.166, 0.103, 0.191 for the three contrasts); the prose/table contradiction disappears (control minus original now includes 0, but that is "not detected", not equivalence); and the "iff" statement is withdrawn — what survives is that the mean differences have opposite signs at the tested mismatch levels, with the η=1 cost robust and the η=0 benefit not established.

Table 2: The first version's repair record, recomputed with correct replication units. At η=0 the 24 rows are 8 units (the target is direction-free); the interval barely excludes 0 and neither p-value is small, so the claim "enforcing equivariance causally improves risk at η=0" is *not* supported by this record. At η>0 the units are (direction, seed) and the cost of enforced invariance is robust.

η	contrast	rows	units	mean diff	95% CI	p_{boot}	p_{flip}
0.0	repaired − control	24	**8**	−0.00093	$[-0.00217, -0.00002]$	0.083	0.125
0.0	repaired − original	24	**8**	−0.00201	$[-0.00394, -0.00021]$	0.034	0.125
0.0	control − original	24	**8**	−0.00107	$[-0.00277, +0.00041]$	0.191	0.320
0.5	repaired − control	24	24	+0.01817	$[+0.01317, +0.02305]$	$< 10^{-4}$	$< 10^{-4}$
0.5	control − original	24	24	−0.00127	$[-0.00546, +0.00289]$	0.549	0.564
1.0	repaired − control	24	24	**+0.04523**	$[+0.03225, +0.05773]$	$< 10^{-4}$	$< 10^{-4}$
1.0	control − original	24	24	−0.00204	$[-0.01346, +0.00903]$	0.721	0.729

Table 3: Floor accounting on the same record, with per-unit floors instead of one quoted number (the "floor" column is the mean over the 8/24 units; the repaired arm's per-unit excess ranges $[-0.0098, +0.0524]$ at η=1). The floor constrains *invariant* predictors only, so the asymmetric arms sit *below* it by fitting the breaking component; the repaired arm sits +0.023 above it (mean over units), which is why "pinned near the floor" is replaced by a number.

η	arm	mean dev risk	floor (mean over units)	excess over floor
0.0	original	0.01511	0.000	+0.01511
0.0	repaired	**0.01311**	0.000	+0.01311
0.0	control	0.01404	0.000	+0.01404
0.5	original	0.23842	0.23710	+0.00132
0.5	repaired	0.25532	0.23710	+0.01822
0.5	control	0.23715	0.23710	+0.00005
1.0	original	0.92796	0.94838	−0.02042
1.0	repaired	0.97114	0.94838	**+0.02276**
1.0	control	**0.92591**	0.94838	−0.02247

What WP-REPAIR-2 decides. With the unit definition enforced and three asymmetric controls, the η=0 picture is: the repaired arm is better than the `original` (−0.00134, Holm p=0.043) and better than the random-direction control (−0.00171, Holm p=0.006), but the gate-count-matched duplicate control closes most of that gap (−0.00060, CI $[-0.00135, +0.00019]$), and the operator-matched control is indistinguishable from the original (+0.00003) — which is the validation that the matching works. The defensible statement is therefore: *under this training procedure, enforcing equivariance improves risk over the asymmetric architectures we tried by an amount whose symmetry-specific part is smaller than our resolution*, and the honest headline is the asymmetry: the cost at η>0 is 60× the size of the η=0 benefit and is detected at every budget. Table 5 shows why: the repaired arm's predictor is invariant to roundoff ($\mathbb{E}[(f - \mathcal{T}_G f)^2] \sim 10^{-33}$) while the asymmetric arms keep $O(10^{-3})$ of breaking component, which is worth 0.02 of risk at η=1.

Budget sensitivity. Repeating the principal contrast at 600 and 1800 Adam steps (fewer units: 6 at η=0, 12 at η>0) leaves the η=0 repaired-versus-duplicate-control difference undetected at every budget (−0.00135, CI $[-0.00316, +0.00006]$ at 600; −0.00111, CI $[-0.00288, +0.00014]$ at 1800),

Table 4: WP-REPAIR-2 with the unit definition enforced by design: η=0 is emitted once (the target is direction-free) with 24 optimiser seeds, η>0 has 8 directions × 24 seeds = 192 units. Holm is applied within the η=0 and η=1 families. Reading: the repaired arm does beat the `original` and the random-direction control at η=0, but the *gate-count-matched duplicate* control closes most of that gap, so the symmetry-specific part (−0.0006) is not established. At η>0 the cost of enforced invariance is robust and grows with the optimiser budget.

η	contrast	units	mean diff	95% CI	p_{boot}	p_{flip}	p_{Holm}
0.0	repaired − control_dup	24	−0.00060	[−0.00135, +0.00019]	0.123	0.150	0.246
0.0	repaired − control_rand	24	**−0.00171**	[−0.00276, −0.00080]	0.0012	0.0004	**0.006**
0.0	repaired − original	24	**−0.00134**	[−0.00237, −0.00031]	0.011	0.018	**0.043**
0.0	control_dup − original	24	−0.00074	[−0.00166, +0.00002]	0.082	0.133	0.246
0.0	control_matched − original	24	+0.00003	[−0.00033, +0.00053]	0.875	0.937	0.875
0.5	repaired − control_dup	192	+0.01142	[+0.00958, +0.01329]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
0.5	repaired − original	192	+0.00879	[+0.00719, +0.01043]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
0.5	control_dup − original	192	−0.00263	[−0.00457, −0.00069]	0.0083	0.0081	0.017
0.5	control_matched − original	192	−0.00043	[−0.00124, +0.00032]	0.276	0.295	0.276
1.0	repaired − control_dup	192	**+0.03605**	[+0.03134, +0.04087]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
1.0	repaired − control_rand	192	+0.02797	[+0.02296, +0.03316]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
1.0	repaired − original	192	+0.02526	[+0.02135, +0.02920]	$< 10^{-4}$	$< 10^{-5}$	$< 10^{-3}$
1.0	control_dup − original	192	−0.01080	[−0.01545, −0.00620]	$< 10^{-4}$	2×10^{-5}	$< 10^{-3}$
1.0	control_matched − original	192	−0.00234	[−0.00439, −0.00043]	0.021	0.021	0.021

while the η=1 cost of enforced invariance *grows* with the budget (+0.036 at 200, +0.051 at 600, +0.057 at 1800 steps, all intervals excluding 0): more optimisation makes the asymmetric arms better at fitting the breaking component, which is the mechanism Proposition 1 predicts.

7.2 Search: the omitted endpoint is the one that works

Table 6 is the substance of referee point 2. The first version reported `verify-repair` versus `verify` on the *unaveraged* candidate risk and concluded that an averaging benefit seen in a 3-seed pilot "did not survive" confirmation. Measured on the deployed predictor, the averaging effect on graphs is −0.0486 with an interval excluding 0 (sign-flip p=0.016), and the deployed-pipeline contrast moves from +0.0009 (candidate endpoint) to −0.0274 (deployed endpoint). The primary hypothesis — that verifier *feedback* improves selected risk — remains undetected on both families, and neither interval is inside the prespecified equivalence margin, so we do not claim equivalence either. On the deployed endpoint the repair arm does beat the fitness arm (−0.04158, CI [−0.08572, −0.00566], shifted-bootstrap p=0.041), while on the candidate endpoint the same pair is undetected (−0.01322): the endpoint correction is what makes the arm comparison meaningful.

Coverage. On `graphs_n6`, 7 of 12 runs attempted a repair and all 7 fired; on `ring_n6`, 4 attempted and 0 fired (the label-preservation gate rejected them). E-S1 is therefore a per-protocol estimate; the arm-level intention-to-treat question is answered by E-S2, which is underpowered at 12 seeds.

The tool, not the LLM. Applying the same declared-group averaging to every arm's selected predictor (E-S5b) lowers development risk for *all* arms, including the non-LLM ones: graphs −0.0639

Table 5: WP-REPAIR-2 floor accounting on the orbit-completed distribution (exact finite-orbit floor, uniform weights over the 12 group images), with the measured predictor invariance $\mathbb{E}[(f - \mathcal{T}_G f)^2]$ and the mean channel residual. The repaired arm is exactly invariant (residual and asymmetry at roundoff, as proved in Appendix A); the asymmetric arms sit below the floor at η=1 by fitting the breaking component.

η	arm	dev risk	orbit risk	exact floor	excess	$\mathbb{E}[(f - \mathcal{T}_G f)^2]$
0.0	original	0.01471	0.01473	0.00000	+0.01473	1.4×10^{-4}
0.0	repaired	**0.01337**	0.01337	0.00000	+0.01337	9×10^{-34}
0.0	control_dup	0.01397	0.01399	0.00000	+0.01399	4×10^{-5}
0.0	control_matched	0.01475	0.01477	0.00000	+0.01477	1.8×10^{-4}
0.0	control_rand	0.01508	0.01509	0.00000	+0.01509	1.0×10^{-4}
0.5	original	0.24164	0.24612	0.24021	+0.00591	1.4×10^{-3}
0.5	repaired	0.25042	0.25587	0.24021	+0.01565	1.2×10^{-33}
0.5	control_dup	**0.23900**	0.24289	0.24021	+0.00268	2.0×10^{-3}
1.0	original	0.93566	0.95210	0.96085	−0.00875	1.8×10^{-3}
1.0	repaired	0.96091	0.98029	0.96085	**+0.01944**	1.6×10^{-33}
1.0	control_dup	**0.92486**	0.94043	0.96085	−0.02042	2.8×10^{-3}
1.0	control_matched	0.93332	0.95003	0.96085	−0.01082	2.0×10^{-3}
1.0	control_rand	0.93294	0.95224	0.96085	−0.00861	2.0×10^{-3}

Table 6: E-S3 (primary) and E-S1 (the endpoint the first analysis omitted). E-S1 is coverage-conditional: on `ring_n6` the repair never fired, so there is no E-S1 estimate there at all — an absence the first version did not report.

estimand	family	n	mean diff	95% CI	verdict
E-S3 verify − fitness	graphs	12	−0.01414	[−0.05855, +0.02837]	not detected
E-S3 verify − fitness	ring	12	+0.00058	[−0.00024, +0.00140]	not detected
E-S1 averaged − candidate	graphs	**7/12**	**−0.04862**	[−0.07083, −0.02839]	averaged better
E-S1 averaged − candidate	ring	**0/12**	no data (4 attempted, 0 fired)		
E-S2 deployed verify-repair − verify	graphs	12	−0.02744	[−0.05401, +0.00519]	not detected
E-S2 deployed verify-repair − fitness	graphs	12	−0.04158	[−0.08572, −0.00566]	repair arm better
E-S2 deployed verify-repair − verify	ring	12	−0.00046	[−0.00127, +0.00045]	not detected

(`llm-fitness`) to −0.1114 (`random`); ring −0.0031 to −0.0039; all intervals exclude 0. Whatever the averaging buys, it does not require an LLM to buy it.

7.3 Verification does not identify the group

For every proposal that carried a hypothesis we recomputed, on the trusted side, whether its group equals the task's group and whether it passes the label-preservation gate that the verifier uses. On `graphs_n6`, 299 proposals carried a hypothesis: 10 carried S_6, and 289 wrong-group proposals *passed* the gate. On `ring_n6`, 192 carried a hypothesis: 0 carried D_6, and 7 passed.

This is not an implementation bug; it is the test's logic. Label preservation tests $y(gx) = y(x)$ on sampled rows for the *proposed* group. If the true label symmetry is G and the proposal is a subgroup $H \leq G$, every element of H preserves labels by construction, so the gate accepts. It is a *soundness filter* (it can refute a group that visibly breaks labels, as on the ring where 185/192 are rejected)

Table 7: Breaking-target selectivity test (development risk on each run's own validation split; refits reproduce logged risks to $\leq 7 \times 10^{-18}$). The gate's *accept* rate tracks label preservation (8 of 348 hypotheses pass here, against 299 of 299 on the invariant graphs target), so the filter works; identification still fails (0 of 348 carry D_6); repair never fires; and no arm ordering is detectable.

arm	dev risk	accuracy	hyp.	pass gate	= task group	repair fired
random	0.04122	0.9122	69	0	0	—
mutation-ea	0.04123	0.9153	79	0	0	—
llm-fitness	0.04049	0.9118	56	0	0	—
llm-verify	0.03699	0.9163	73	4	0	—
llm-verify-repair	0.04106	0.9170	71	4	0	0 of 6

paired contrasts (family E, Holm-adjusted)
verify − fitness -0.00350 $[-0.00963, +0.00198]$, p_{Holm}=0.74
verify-repair − verify $+0.00407$ $[-0.00069, +0.00984]$, p_{Holm}=0.67
verify-repair − fitness $+0.00058$ $[-0.00535, +0.00649]$, p_{Holm}=1.00
verify − random -0.00423 $[-0.01027, +0.00135]$, p_{Holm}=0.67
mutation-ea − random $+0.00001$ $[-0.00280, +0.00311]$, p_{Holm}=1.00

and not an *identification* test. The behavioural consequence is exactly what the search results show: repair fires on candidates whose group is wrong, and the averaged predictor still improves, because averaging over a subgroup of the true symmetry still removes variance. Establishing identification would need a different instrument (counterexample search over the group, or a symmetry-breaking probe distribution), which we did not build.

7.4 Does the verifier's effect track the target's symmetry? (Addendum A1)

The selectivity test was declared in the protocol before it was run: all five arms on `ring_n6` with target `alpha_1` (the declared group is *not* a label symmetry of this target), 12 fresh paired seeds, its own Holm family, and the prediction stated in advance — if the verifier's benefit came from pushing the search toward the declared group, it must turn negative when that group is wrong.

Three things follow. The verifier's *filter* is selective: it accepts 100% of hypotheses when the target is invariant and 2.3% when it is not, which is the behaviour its definition promises. The verifier's *identification* is not improved: no proposal carries the task's group, and the repair step never fires. And the verifier's *effect on search* is undetectable in either direction (largest absolute mean difference 0.0042, all Holm $p \geq 0.67$), so the pre-declared prediction is not testable at this resolution: with the symmetry removed, verification neither measurably helped nor measurably hurt. That is consistent with the primary null — the feedback channel rarely changes the selected candidate — and inconsistent with any large "verification helps" story. For reference, moving from the invariant to the breaking target costs every arm ≈ 0.031 of risk (0.0065–0.0102 to 0.037–0.041), and no arm detects the difference.

7.5 Baselines the LLM story has to beat

At the same fit budget (120 steps, 256 rows, 12 paired seeds), the hand-built equivariant architecture (bidirectional CRZ, measured channel residual 4×10^{-15}) reaches development risk 0.00055 on `ring_n6`; every search arm is worse, by $+0.0060$ (`llm-fitness`, CI $[+0.0055, +0.0064]$) to $+0.0097$ (`mutation-ea`), all $p < 10^{-4}$. On `graphs_n6` the same construction is *worse* than every arm (0.667

Table 8: E-S7: sealed-test risk of the frozen 120-run candidates, refitted deterministically (reproducing logged development risk to 1.7×10^{-18}) and scored once. The sealed split reproduces the development pattern: on the ring the LLM arms beat the non-LLM arms and the verifier-versus-fitness difference is undetected; on graphs no arm difference is detected and every arm sits far above the 0.351 edge-count rule.

family	arm	dev risk	sealed-test risk	paired vs `random` on test
ring	random	0.01013	0.00995	—
ring	mutation-ea	0.01022	0.01025	+0.00030 [−0.00009, +0.00090]
ring	llm-fitness	0.00651	0.00639	**−0.00356** [−0.00469, −0.00239]
ring	llm-verify	0.00709	0.00711	**−0.00284** [−0.00442, −0.00117]
ring	llm-verify-repair	0.00664	0.00697	**−0.00298** [−0.00452, −0.00154]
graphs	random	0.53782	0.48704	—
graphs	mutation-ea	0.53930	0.49753	+0.01049 [−0.01082, +0.03816]
graphs	llm-fitness	0.56572	0.51806	+0.03102 [−0.04167, +0.09514]
graphs	llm-verify	0.55158	0.49272	+0.00568 [−0.04052, +0.05050]
graphs	llm-verify-repair	0.55250	0.50674	+0.01970 [−0.00448, +0.05037]

vs 0.538–0.566): an exactly equivariant three-parameter block cannot express connectedness, which is the two-sided lesson of two-sided lesson of [4]. The classical edge-count rule (threshold fitted on training rows only, k=6) reaches development risk 0.351 and accuracy 0.912 on graphs, beating every arm; the best constant predictor reaches 0.657; the invariant oracle on the ring is 0 by construction. Density stratification (E-S5) puts the arms' error where the shortcut lives: in the 0–5 edge band the five arms' mean risk is 1.439 against 0.167 in the 8–9 band, and the edge-count rule's band risks are 1.13 (0–5), 0.13 (6–7), 0.00 (8–9) and 0.05 (10–11) — the same profile, which is why we describe the graphs task as shortcut-dominated rather than claiming the shortcut *caused* the null without an intervention that changes arm ordering.

7.6 One-shot sealed evaluation of the search arms (E-S7)

Every development-selected candidate was refitted deterministically (reproducing its logged development risk to 1.7×10^{-18}) and scored once on the sealed split, with the file digests recorded before and after (`research/HOLDOUT_ACCESS.json`).

The two prespecified contrasts of interest behave as on development: `llm-verify` − `llm-fitness` on the ring is +0.00072 (CI [−0.00039, +0.00186], not detected) and on graphs −0.02534 (CI [−0.11142, +0.05299], not detected), while `llm-fitness` − `mutation-ea` on the ring is −0.00386 (CI [−0.00491, −0.00277]). Test risks are uniformly a little below development risks on graphs (the sealed split is easier), so the level of the development numbers is not an artifact of the split — but no arm difference on graphs survives the move to sealed data, which is the third time this section has reported a development effect that does not materialise.

7.7 Untouched-input evaluation of the repair arms (E-R4')

The first one-shot attempt at this evaluation was *invalid* and is reported as such: the evaluator scored every arm against a single invariant label array per split, so at η>0 it answered a different question (the tell was a risk of 0.019 at η=1 where the invariant floor is ≈ 1). Protocol Addendum A4 records the defect, keeps the two ring splits it had already read consumed, and replaces them with

Table 9: E-R4' on untouched inputs (4 breaking directions × 8 optimiser seeds: 8 units at η=0, 32 at η>0), next to the same unit subset on development rows. The untouched data reproduce the WP-REPAIR-2 reading: the repaired arm beats the original architecture, the gate-count-matched duplicate control is *not* beaten at η=0, and enforcing invariance costs 0.046 of risk at η=1. Magnitudes differ slightly from Table 4 because this is a subset of directions and seeds.

	contrast	development (same units)		untouched inputs	
η		mean diff	95% CI	mean diff	95% CI
0.0	repaired − control_dup	−0.00083	[−0.00225, +0.00024]	−0.00057	[−0.00176, +0.00028]
0.0	repaired − original	**−0.00233**	[−0.00417, −0.00057]	**−0.00174**	[−0.00309, −0.00051]
0.0	control_dup − original	−0.00150	[−0.00351, +0.00024]	−0.00117	[−0.00279, +0.00015]
0.5	repaired − control_dup	+0.01757	[+0.01326, +0.02180]	+0.01551	[+0.01106, +0.02011]
0.5	repaired − original	+0.01265	[+0.00875, +0.01658]	+0.01153	[+0.00804, +0.01501]
0.5	control_dup − original	−0.00492	[−0.00955, −0.00033]	−0.00399	[−0.00853, +0.00043]
1.0	repaired − control_dup	+0.05138	[+0.04083, +0.06123]	**+0.04629**	[+0.03613, +0.05670]
1.0	repaired − original	+0.03440	[+0.02521, +0.04384]	+0.03067	[+0.02227, +0.03906]
1.0	control_dup − original	−0.01697	[−0.02978, −0.00450]	−0.01561	[−0.02814, −0.00358]

an **inputs-only** holdout (seed 20261204, 4000 rows, zero dihedral-orbit overlap with development or any earlier test split, sha256 recorded, read once), evaluated only after a gate that refits every WP-REPAIR-2 unit and must reproduce the experiment's own development risk exactly (`216/216` compared, max difference 0.0).

The untouched-input result is the cleanest statement this paper can make about repair: on data no part of the pipeline touched, enforcing equivariance buys nothing over a gate-matched asymmetric control when the target is symmetric, and costs 4.6% of the target variance when it is not. The file digests before and after the read are identical (`research/HOLDOUT_ACCESS_v4.json`), and the evaluator now refuses to open the two splits consumed by the defective run.

8 Discussion

What the evidence supports. (a) Enforcing equivariance has a large, robust *cost* when the target breaks the symmetry (+0.03605, CI [+0.03134, +0.04087], 192 units, rising to +0.057 at 1800 optimiser steps), and at η=0 a benefit that the gate-count-matched control removes: the repaired arm beats the original architecture but not the duplicate-gate control, so no symmetry-specific benefit is established. (b) The analytic repair — inference-time averaging over a declared group — is the only intervention in this study that improves the deployed predictor consistently, and it does so for non-LLM pipelines too. (c) Executable verification as implemented does not identify symmetry groups, for a reason visible from group theory before the experiment. (d) At matched budgets, constructive equivariant design and a one-threshold classical rule beat LLM search on these two tasks.

What that implies for the "verification helps automated science" thesis. Verification helped in exactly one place: turning a wrong-but-implemented circuit into a better predictor by post-processing it with a group we supplied. It did not help the model *find* the group, and the feedback channel we built (residual plus label report) did not move selected risk. We therefore

reposition the claim: in this setting the value of an executable verifier is as a precondition for an analytic repair, not as a teacher.

Relation to the pilot. A 48-run pilot (3 seeds, 3 treatments) preceded the 120-run study and is retained as exploratory. Its variance check set the confirmatory seed count. Dated git evidence for the pilot's protocol is not available (Appendix C), so we describe the search study as protocol-consistent rather than verifiably pre-registered; the repair-2, dev-analysis and one-shot holdout chain is verifiably pre-registered (protocol commit `ecd1be72`, 16:20:02Z, precedes job submissions at 17:10Z and 17:21Z).

9 Limitations

Small dense statevector simulation; two six-qubit development families plus one sealed ring holdout, with `tictactoe_n9` left sealed and unclaimed because dense n=9 simulation exceeds this revision's budget; a single model (*Qwen3.8-Flash-Next*), so no model comparison is claimed; budget B=12; three (now eight) breaking directions, which supports within-benchmark statements only; equal gate counts do not make two circuit families equivalent in expressivity or optimisation geometry, so no symmetry-only attribution is offered; the verifier's label gate has no power against subgroups of the true label symmetry; the model serving alias exposes no immutable checkpoint revision, so reproducibility of the LLM component is temporal rather than bit-level; and no quantum-advantage claim is made or implied.

10 Reproducibility and integrity

All splits are orbit-disjoint; the sealed splits were opened once for outcomes after the protocol freeze, with a metadata-only access incident recorded in `research/PROVENANCE.md` and remedied by a fresh holdout generated after the freeze; every result JSON records seeds, module versions, git commit and host (with two recorded defects in the older records); analysis code, run records, dataset manifests, the frozen protocol, the bibliography audit and the per-table commands are released with the paper. Negative and null results are reported as "not detected" with their intervals, and equivalence is claimed only against a prespecified margin.

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A CRZ algebra behind the arms

With the simulator's convention (q_0 most significant, $\mathrm{CRZ}_{ij}(b) = |0\rangle\langle 0|_i \otimes I_j + |1\rangle\langle 1|_i \otimes \mathrm{RZ}(b)_j$),

$$\mathrm{CRZ}_{ij}(b)\,\mathrm{CRZ}_{ji}(b) = \exp\!\big(-\tfrac{ib}{4}(Z_i + Z_j)\big)\,\exp\!\big(+\tfrac{ib}{2} Z_i Z_j\big), \tag{4}$$

$$\mathrm{CRZ}_{ij}(b)^2 = \mathrm{CRZ}_{ij}(2b) = \exp\!\big(-\tfrac{ib}{2} Z_j\big)\,\exp\!\big(+\tfrac{ib}{2} Z_i Z_j\big). \tag{5}$$

Both are verified numerically to 10^{-13} in `research/experiments/repair/test_repair2.py`. Equation (4) is the analytic reason the repaired arm is equivariant: the ZZ term is manifestly symmetric and the field term is a permutation-symmetric product of identical single-qubit rotations, so the block commutes with *every* permutation, not only D_6; the measured residual (4×10^{-16}) is a check, not the proof. Equation (5) is the analytic reason the gate-count-matched control is not: its field term hits only the target wire, so local fields are non-uniform. It also shows the control is the `original` interaction at angle $2b$, which is why an operator-matched control (`control_matched`, initial b halved) is needed and why the two controls are reported separately.

B Reproduction

Tests (all pass):

```
python3 -m unittest discover -s research/core      -p 'test_*.py'
python3 -m unittest discover -s research/experiments/repair -p 'test_*.py'
python3 -m unittest discover -s research/search    -p 'test_*.py'
```

Experiments and analyses (one command per table):

```
python3 -m research.experiments.repair.repair_experiment2 --workers 48 \
  --out research/experiments/repair/repair2_results.json
python3 -m research.experiments.repair.analyze_repair \
  --in  research/experiments/repair/repair_results.json \
  --out research/experiments/repair/repair_v1_reanalysis.json
```

```
python3 -m research.experiments.repair.analyze_repair \
  --in  research/experiments/repair/repair2_results.json \
  --out research/experiments/repair/repair2_analysis.json
python3 -m research.search.analyze2 --dir logs/runs_confirm \
  --out research/search/confirm_analysis2.json --label confirmatory
python3 -m research.search.dev_analysis --dir logs/runs_confirm \
  --out research/search/dev_analysis.json --workers 24
python3 -m research.search.analyze_breaking --dir logs/runs_breaking \
  --out research/search/breaking_target.json
python3 -m research.holdout_oneshot --splits dev --skip-search \
  --selfcheck research/experiments/repair/repair2_results.json   # gate, no sealed read
python3 -m research.holdout_oneshot --splits fresh_v3 --skip-search \
  --selfcheck research/experiments/repair/repair2_results.json \
  --out research/HOLDOUT_RESULTS_v3.json \
  --access-log research/HOLDOUT_ACCESS_v3.json   # ONE shot on v3 only
python3 -m research.compile_results2 --out research/results_summary2.md
```

Stack: `Python/3.12.3`, `SciPy-bundle/2024.05` (NumPy 1.26.4, SciPy 1.13.1); LLM arms additionally `module load ycrc-llm/1.0`. CPU batch jobs run on the cluster's `day` partition with `OPENBLAS_NUM_THREADS=1`; the work is CPU-only, so no GPU partition is used (holding a GPU idle to satisfy old protocol wording would violate cluster utilization policy).

C Provenance

Model identity: the alias `ycrc-local` resolves to *Qwen3.8-Flash-Next*, root `/apps/software/openwebui_models/Qwe` served by vLLM with `max_model_len` 262144; requests send `model` and `reasoning_effort="none"` only (no temperature, top_p or max_tokens), with a 180 s timeout and up to 8 retries (none needed: 584/584 calls succeeded on the first attempt, mean latency 2.75 s). Budgets for the 120-run study: 584 calls, 1,294,840 tokens (829,452 prompt / 465,388 completion), 1448 proposals of which 8 were invalid and retained, 18.1 core-hours. The gateway publishes no immutable checkpoint revision, so the pin is alias + root path + module version + date.

Two provenance defects in our own earlier records, recorded rather than repaired silently: (1) the run JSONs report `git_commit=a37fc2d6`, which does not contain the code that produced them (it was uncommitted at run time); (2) the plan document that claims to have been frozen before any outcome was seen first appears in git *after* the pilot and confirmatory runs, so the search study is described as protocol-consistent, not verifiably pre-registered. Full dated chain, sealed-file digests and the access incident: `research/PROVENANCE.md`.